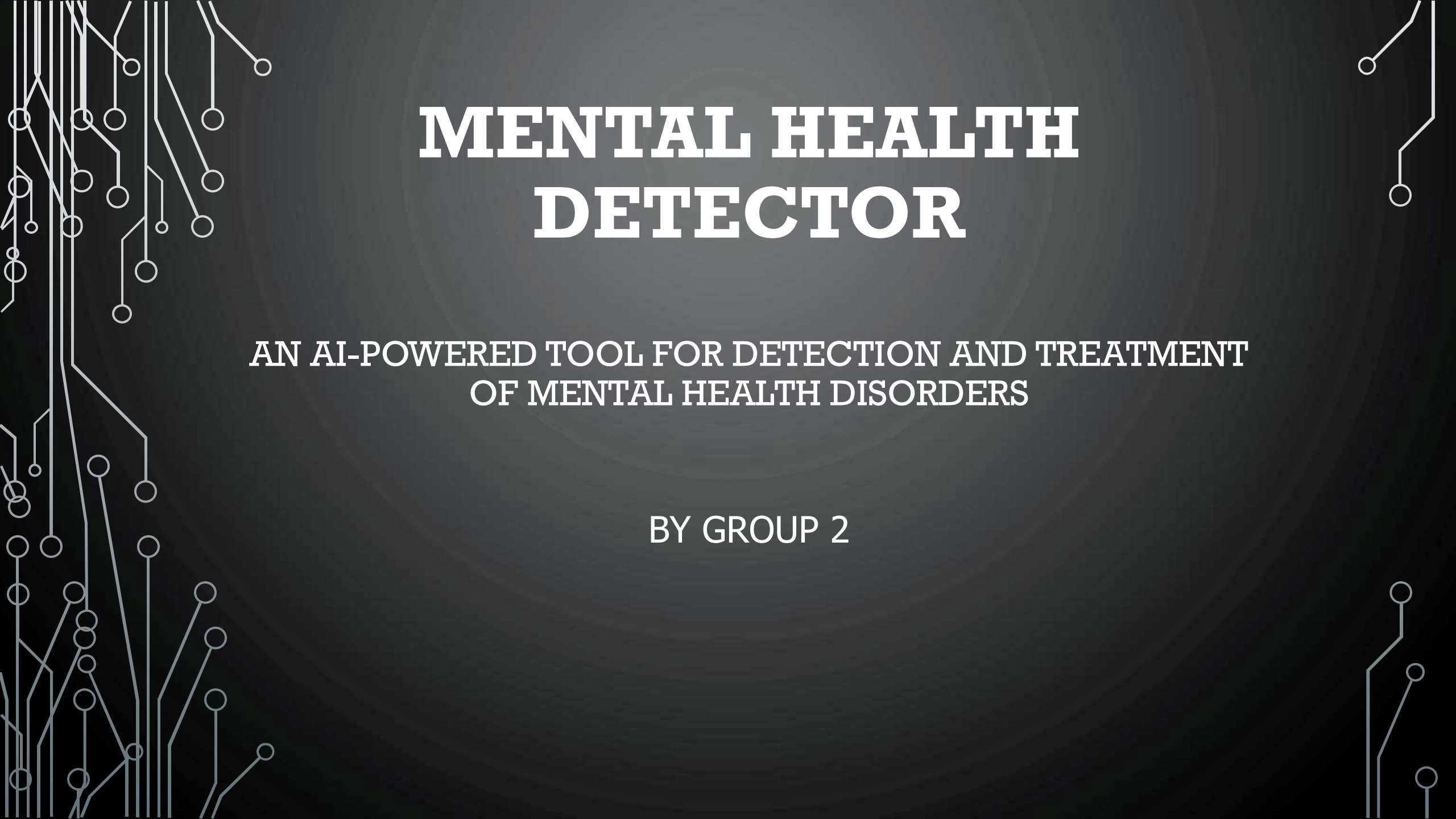


The background features a dark gray gradient with faint, concentric circular patterns. White circuit-like lines, consisting of straight segments and small circles, are positioned along the left and right edges, resembling a stylized circuit board or neural network.

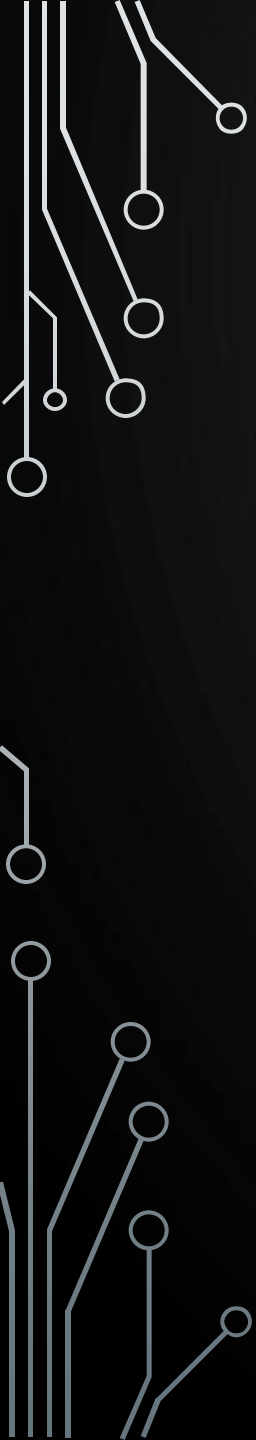
MENTAL HEALTH DETECTOR

**AN AI-POWERED TOOL FOR DETECTION AND TREATMENT
OF MENTAL HEALTH DISORDERS**

BY GROUP 2

INTRODUCTION

- This project delivers an integrated mental health prediction and risk assessment tool, combining natural language processing for symptom-based disorder identification with machine learning for lifestyle-based risk evaluation. The system is designed for use by individuals, clinicians, and mental health practitioners, providing early detection, risk scoring, and actionable recommendations. The platform supports two complementary functionalities:
 1. Signs-Based Disorder Prediction – Users provide textual descriptions of symptoms, which are transformed into vector embeddings and classified into likely mental disorders. The model then returns relevant self-care and professional recommendations.
 2. Lifestyle-Based Risk Assessment – Users input structured lifestyle and demographic data, and the model predicts both the likelihood and severity of mental health risk.
- By combining these approaches, the application aims to provide holistic mental health evaluation, increasing awareness, promoting early intervention, and supporting informed decision-making.



BUSINESS UNDERSTANDING; PROBLEM STATEMENT

- **BACKGROUND**

Mental health significantly impacts daily life, yet many people—especially in low- and middle-income regions—face barriers to care, including stigma and lack of access. With over 1 in 8 people globally affected by mental health disorders, there is a pressing need for early and accessible support. AI can help fill this gap by detecting emotional distress through text, offering supportive feedback, and encouraging self-reflection. These tools not only aid individuals but also assist therapists by tracking emotional trends and enhancing patient care, making AI a valuable partner in mental health support.

- **PROBLEM STATEMENT**

Many people struggle with mental health issues but face barriers like cost, stigma, and limited access to therapists. Emotional distress often goes unaddressed, and even those in care lack tools for tracking emotions between sessions. There's a growing need for accessible, private tools that support self-reflection and provide valuable insights to therapists.



PROJECT DESCRIPTION; OBJECTIVES

1. Analysis-based objectives

- Analyze user-generated text to uncover emotional and linguistic patterns associated with mental health conditions such as anxiety, depression, and burnout.
- Identify trends and frequencies of symptoms that may signal the onset or escalation of mental health challenges.
- Categorize emotional expression into broad mental health categories using historical input data.
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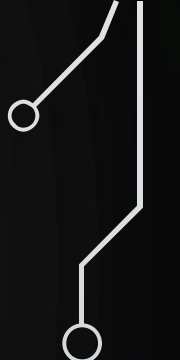
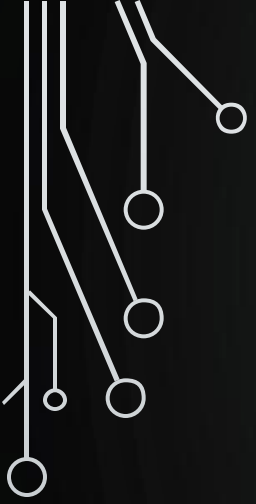
2. Feature engineering objectives

- Preprocess textual data through tokenization, lemmatization, and removal of stop words to ensure clean input for model training.
 1. Extract relevant features such as:
 2. **Emotional tone** (positive, negative, neutral)
 3. **Symptom-related keywords and phrases**
 4. **Frequency and persistence** of emotional states
 5. **Time-aware features** for tracking changes over time

PROJECT DESCRIPTION; OBJECTIVES

3. Modeling Objectives

- Develop and fine-tune a multi-label NLP model that can detect multiple co-occurring mental health conditions from text input.
- Integrate the model into an interactive **chatbot system** designed to:
 - Deliver empathetic, condition-aware responses
 - Encourage users to perform daily emotional check-ins or journaling
 - Provide coping strategies and motivational content
 - Assess distress levels (mild, moderate, severe) to tailor response tone and urgency
- Evaluate model performance using relevant multi-label metrics including **Hamming Loss**, **Subset Accuracy**, and **F1-score**.



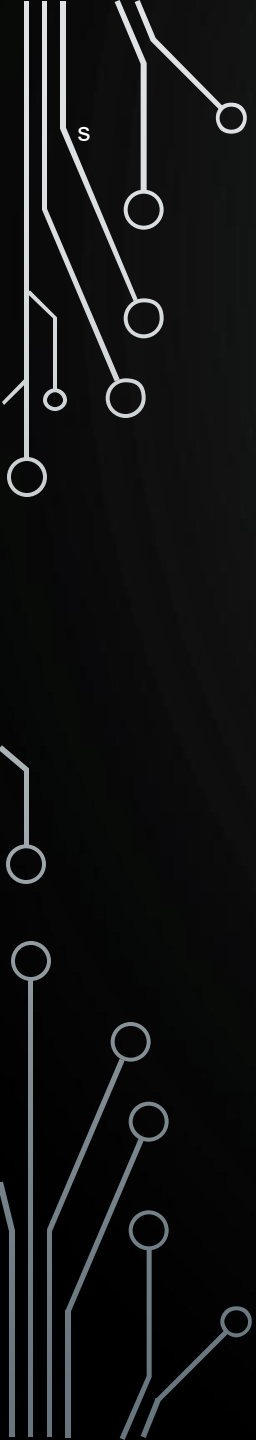
PROJECT DESCRIPTION; DATA DESCRIPTION

1. Mental Health Content (mentalhealth.org.uk):

- Signs, symptoms, and emotional indicators
- Self-care and professional help recommendations
- Used to train emotional tone detection and generate supportive chatbot responses

2. Demographic and behavioral data:

- Features: age, gender, occupation, stress levels, physical activity
 - Includes user input and mental health condition labels
 - Used to train the multi-label classification model
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DATA PREPARATION

1. Mental health content

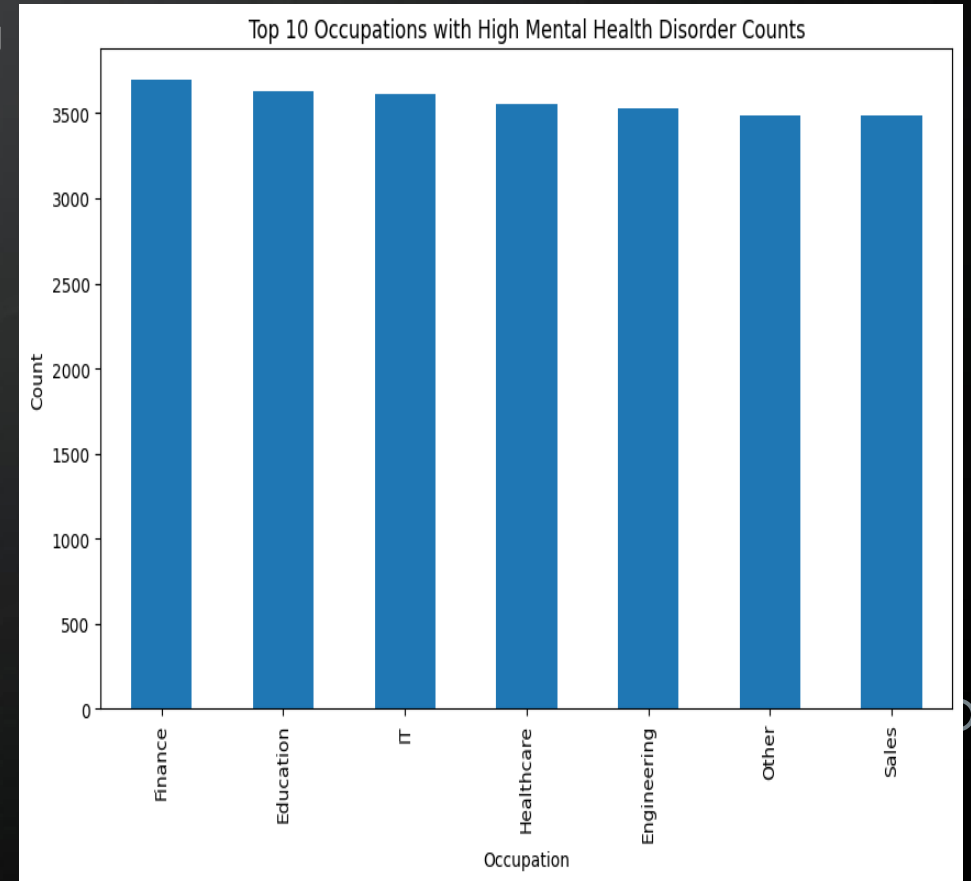
- Focused on descriptive mental health content: Signs, symptoms, coping strategies, and recommendations
- No missing values
- Processed using **SentenceBERT** to generate embeddings for classification and chatbot response modeling

2. Demographic and behavioral data

- Included structured features like: Age, gender, occupation, diet quality, smoking habit, alcohol consumption, sleep hours, physical activity hours, stress level, social media usage, consultation history, medication usage, work hours.
 - Missing values in **severity** column handled using **mode imputation**
 - Used to train the **CatBoost** model for multi-label classification and risk prediction
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EXPLANATORY DATA ANALYSYS

- Before continuing to modeling, we tried some EDA to uncover data patterns. And we found;
 - The majority of users are aged between 20 and 35, showing mental health engagement is stronger among young adults.
 - Gender distribution is fairly balanced across the dataset.
 - High stress levels are commonly reported, indicating it's a key factor for mental health issues.
 - Occupation-wise, mental health disorder counts are highest in fields like **Finance, Education, IT, and Healthcare**, suggesting certain professions may be more affected by work-related mental strain.



MODELLING

1. Text-based modeling

- Used **SentenceBERT** to convert user input (symptoms, journal entries) into semantic embeddings
- Trained a **multi-label classifier** to detect conditions like **depression**, **anxiety**, and **burnout**
- Integrated into a chatbot that responds **empathetically** and offers **condition-specific suggestions**

2. Structured data modeling

- Used a stacked model of Gradient Boosting and CatBoost Classifier to train 13 behavioral and demographic features
- Target: Multi-label classification of conditions + severity prediction (mild, moderate, severe)
- Handled missing values using **mode imputation** (severity column)

MODEL INTERPRETABILITY AND EVALUATION

1. Text-based model (sentenceBERT + multi-label classifier)

- **Goal:** Detect mental health conditions (e.g., anxiety, depression, burnout) from symptom descriptions
- **Evaluation Metrics:**
 - **Hamming Loss:**
 - **Subset Accuracy: 81%**
 - **Macro F1-score.**

2. Structured data model (CatBoostClassifier)

- **Models:** We used a stacked model of catboost and lightgbm
- **Goal:** Predict mental health conditions & severity from lifestyle features
- **Evaluation metrics:**
 - **Accuracy: 83%**
 - **Macro F1-score.**
 - **Weighted f1-score.**

DEPLOYMENT

- **Platform:**

1. Streamlit - Streamlit let us run the api locally.
2. API - We built a FastAPI backend for scalable inference.
3. Ngrok - ngrok will let us deploy to the internet for testing.

- **Modes :**

- *Signs-Based* – NLP-powered prediction from user-described symptoms
- *Lifestyle-Based* – Structured data prediction from age, stress, sleep, etc.

- **Model Architecture**

1. `Model_nlp.pkl`
 - SentenceTransformer for symptom encoding
 - Pre-embedded disorder vectors
 - Recommendation DataFrame (self-care, professional, other)
2. `Model_structured.pkl`
 - Risk prediction model (binary classifier)
 - Severity level model (multi-class classifier)
 - Label encoders for categorical inputs

SYSTEM DESIGN

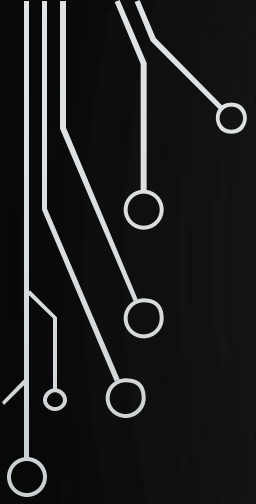
Prediction flow

1. Signs-based flow

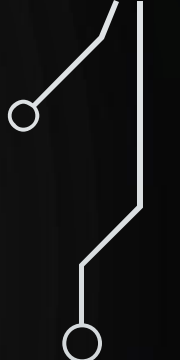
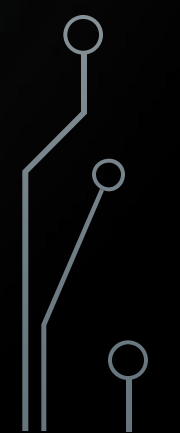
- User inputs text description of symptoms
- Text is embedded using SentenceTransformer
- Compared against disorder embeddings using cosine similarity
- Top 3 disorders shown, with one identified as the top concern
- Recommendations retrieved from the base based on predicted disorder

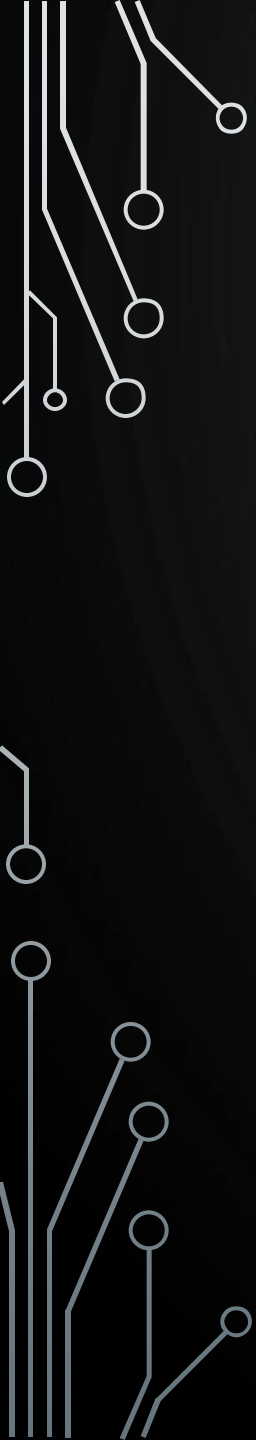
2. Lifestyle-based flow

- User fills in demographic + lifestyle inputs (age, sleep, stress, etc.)
- Inputs are encoded using pre-trained encoders
- Passed into a risk classifier → outputs **Low Risk** or **High Risk**
- If High Risk, severity is predicted (e.g., *mild*, *moderate*, *severe*)
- Then adds recommendations and tips for the disorder
- Also has an option to download csv



CONCLUSION

- The Mental Health Prediction Assistant is a dual-approach system designed for early detection and support in mental health by combining NLP-based symptom analysis with structured lifestyle data.
 - It enables personalized predictions, scalable mental health screening, and tailored recommendations for users.
 - Built with a modular and interpretable architecture, the system is user-friendly and adaptable for real-world use in digital health platforms, awareness campaigns, or personal self-assessment tools.
 - Overall, it serves as a step toward bridging the gap between early signs of mental distress and timely intervention.
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RECOMENDATIONS

1. **Model Performance Improvement** - Enhance the classifier's accuracy by experimenting with more advanced NLP embeddings (e.g., OpenAI embeddings, ClinicalBERT, or BioBERT) for the signs-based model.
 2. **User-Centric Refinement** - Integrate user feedback within the UI to refine predictions and recommendations dynamically over time.
 3. **Integration of Multi-Source Data** - Combine self-reported lifestyle features with wearable health tracker data (sleep quality, heart rate, activity logs) to improve prediction reliability.
 4. **Privacy and Ethics** - Implement robust data encryption and anonymization for all stored or transmitted user inputs.
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NEXT STEPS

- **Continuous Learning System** - Build a feedback loop where user-confirmed diagnoses and lifestyle updates are fed back into the system for continuous model retraining.
- **Multilingual Support** - Expand the system to handle multiple languages to serve a wider audience.
- **Longitudinal Tracking** - Allow users to track changes in their mental health risk scores over time and correlate them with lifestyle adjustments or symptom progression.
- **Regulatory Compliance** - Adapt the platform to comply with HIPAA, GDPR, and other health data regulations depending on target regions for deployment.

The background is a dark gray gradient with several large, faint, concentric circles centered in the middle. In the four corners, there are white line-art patterns resembling circuit boards or neural networks, with lines and small circles connecting them.

THANK YOU.
ANY QUESTIONS?